Deflections of asymmetric deck girders

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Introduction

WSDOT uses three methods of accommodating crown slope and full superelevation in precast deck girder bridges.

1) Girders are constructed with a cross section that is symmetric about the vertical axis and are then installed with their webs perpendicular to the roadway surface.
2) Girders are constructed with a transverse slope in the top flange to accommodate the roadway section and are then installed with their webs plumb. The top flange overhangs are equal length. The girder cross section is asymmetric.
3) Girders are constructed and installed similar to the second method. However, the top flange overhangs are unequal length. The top flange overhang dimensions are chosen such that the center of gravity of the girder is located in a vertical plane passing through the centerline of the web. The girder cross section is asymmetric.

The first two methods are most frequently used. All three methods results in lateral deflection of the girder. For precast deck girder systems that utilize a UHPC closure joint, the alignment of adjacent girder top flanges is crucial. Lateral deflections affect the straightness of the joint as well as the width of the joint along the length of the girder. Figure 1 shows a typical longitudinal joint as designed.

Two items are of concern. Transverse reinforcement extends from the top flange of deck girders into the longitudinal closure joint. This reinforcement must be sufficiently embedded into the joint to develop its tensile capacity. The reinforcement must also have sufficient overlap with the transverse bars extending from the adjacent girder to transfer tensile forces through a non-contact lap splice.

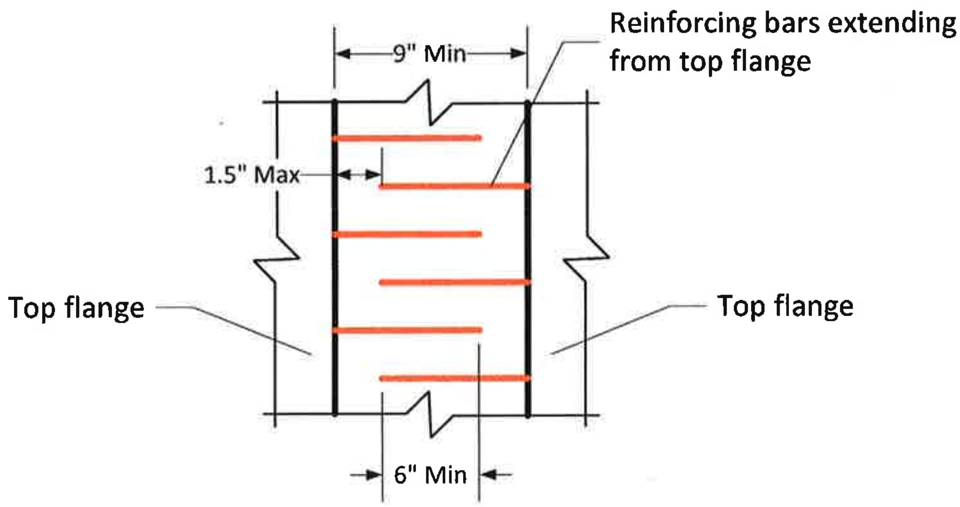


Figure 1 - Longitudinal joint as designed

Girders are not straight when lateral deflections are present. When lateral deflections of adjacent girders are in opposite directions from one another, joint are narrower or wider than designed. Figure 2 and Figure 3 illustrate these conditions. If a joint becomes too narrow, the transverse bars may need shortening in the field to achieve adequate fit up. Joint bars may not develop their full capacity with less development then designed. If a joint becomes too wide, the length of the non-contact lap splice will be reduced and the ability of the joint material to transfer forces across the joint may be compromised.

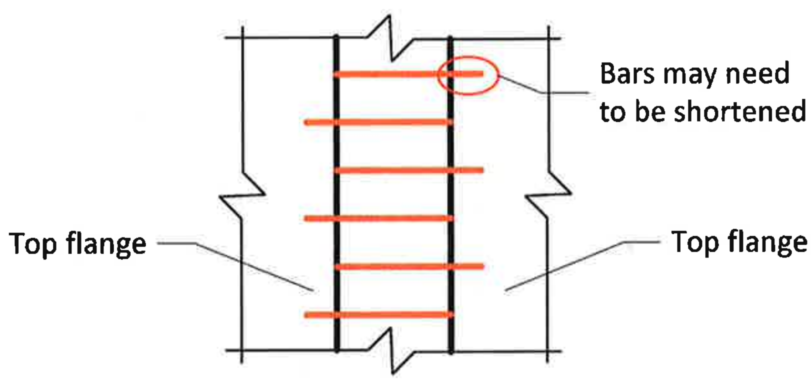


Figure 2 - Narrow longitudinal joint

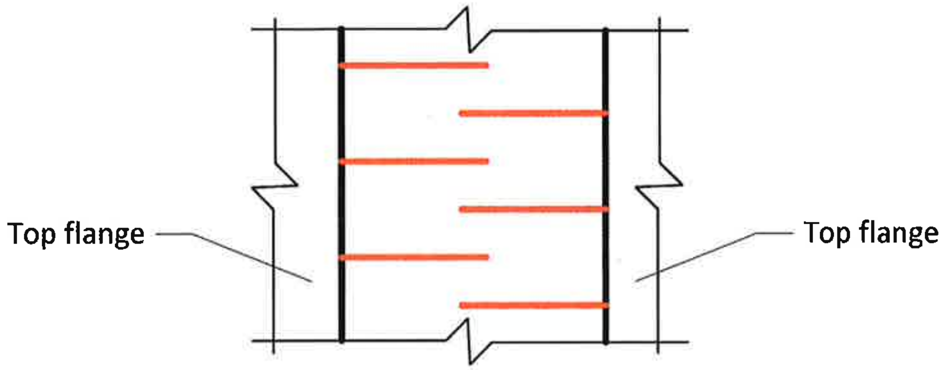


Figure 3 - Wide longitudinal joint

Girders on either side of the crown point will deflect either towards or away from each other. Typically, the crown point is along the centerline of the bridge and the greatest lateral deflection occurs at mid-span. The most poorly aligned location in the longitudinal joint will occur at the location of greatest stress.

Lateral deflection due to prestress will occur when the resultant prestress force is eccentric to the center of gravity of the girder. This is the case with construction methods 2 and 3. WSDOT permits a sweep of 1/8" per 10 ft of girder length not to exceed ½". The lateral deflections due to prestressing may exceed the permissible sweep and girders could potentially be rejected. The rejection would not be warranted because the sweep is due to a design issue, not a fabrication issue. This lateral deflection will contribute to joint fit-up issues and to the instability of the girder during handling. Resolution of these issues is beyond the scope of this study.

This study explores the magnitude of lateral deflections that can occur for the three construction methods listed above. Expressions for the lateral deflections are developed. The lateral deflections are then computed for a precast deck girder bridge with extreme, but theoretically achievable dimension.

Lateral Deflections - Derivation

Figure 4 shows an arbitrary asymmetric section with moments M_x and M_y about the x and y-axes.

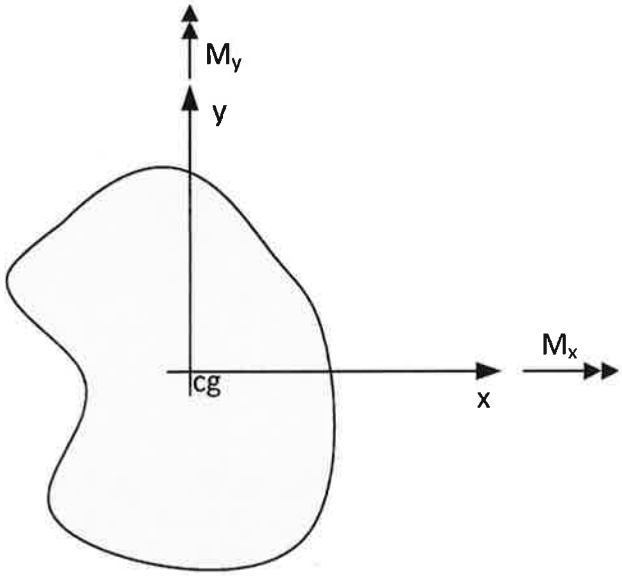


Figure 4 Arbitrary asymmetric section

Let ϕ_x and ϕ_y be the curvatures about the x-axis and y-axis respectively.

Equation 1 relates moments and curvatures.

$$\begin{bmatrix} M_x \\ M_y \end{bmatrix} = E \begin{bmatrix} I_{xx} & I_{xy} \\ I_{xy} & I_{yy} \end{bmatrix} \begin{bmatrix} \phi_x \\ \phi_y \end{bmatrix} \tag{1}$$

Solving Equation 1 for curvature gives,

$$\begin{bmatrix} \phi_x \\ \phi_y \end{bmatrix} = \frac{1}{E\left(I_{xx}I_{yy} - I_{xy}^2\right)} \begin{bmatrix} I_{yy} & -I_{xy} \\ -I_{xy} & I_{xx} \end{bmatrix} \begin{bmatrix} M_x \\ M_y \end{bmatrix} \tag{2}$$

From classical Bernoulli beam theory, curvature is equal to the second derivate of deflection, therefore

$$\phi_x = \frac{d^2y}{dz^2} = \frac{M_x I_{yy} - M_y I_{xy}}{E\left(I_{xx}I_{yy} - I_{xy}^2\right)} \tag{3}$$

$$\phi_y = \frac{d^2x}{dz^2} = \frac{M_y I_{xx} - M_x I_{xy}}{E\left(I_{xx}I_{yy} - I_{xy}^2\right)} \tag{4}$$

Assuming superposition is applicable, the following two cases are independent and the summation of the deflections result in the total deflection.

Case 1 - $M_x \neq 0, M_y = 0$

M_x is caused by gravity loads (self-weight of girder) and prestressing that is eccentric with respect to the x-axis. Substituting $M_y = 0$ into Equations 3 and 4 yields

$$\phi_x = \frac{d^2y}{dz^2} = \frac{M_x I_{yy}}{E(I_{xx}I_{yy} - I_{xy}^2)} \tag{5}$$

$$\phi_y = \frac{d^2x}{dz^2} = \frac{-M_x I_{xy}}{E(I_{xx}I_{yy} - I_{xy}^2)} \tag{6}$$

The deflections in the x and y directions are found by integrating Equations 5 and 6 two times.

$$\Delta_{y1} = \iint \frac{d^2y}{dz^2} = \frac{I_{yy}}{E(I_{xx}I_{yy} - I_{xy}^2)} \iint M_x(z) \tag{7}$$

$$\Delta_{x1} = \iint \frac{d^2x}{dz^2} = \frac{-I_{xy}}{E(I_{xx}I_{yy} - I_{xy}^2)} \iint M_x(z) \tag{8}$$

Re-arranging Equations 7 and 8 gives

$$\iint M_x(z) = \frac{E(I_{xx}I_{yy} - I_{xy}^2)}{I_{yy}} \Delta_{y1} \tag{9}$$

$$\iint M_x(z) = \frac{E(I_{xx}I_{yy} - I_{xy}^2)}{-I_{xy}} \Delta_{x1} \tag{10}$$

Setting Equation 9 equal to Equation 10 and simplifying gives

$$\Delta_{x1} = -\frac{I_{xy}}{I_{yy}} \Delta_{y1} \tag{11}$$

Equation 11 gives the lateral deflection of an asymmetric section as a function of the vertical deflection.

Case 2 - $M_x = 0, M_y \neq 0$

M_y is caused by prestressing that is eccentric with respect to the y-axis. Substituting $M_x = 0$ into Equations 3 and 4 yields

$$\phi_x = \frac{d^2y}{dz^2} = \frac{-M_y I_{xy}}{E(I_{xx}I_{yy} - I_{xy}^2)} \tag{12}$$

$$\phi_y = \frac{d^2x}{dz^2} = \frac{M_y I_{xx}}{E(I_{xx}I_{yy} - I_{xy}^2)} \tag{13}$$

The deflections in the x and y directions are found by integrating Equations 12 and 13 two times.

$$\Delta_{y2} = \iint \frac{d^2y}{dz^2} = \frac{-I_{xy}}{E(I_{xx}I_{yy} - I_{xy}^2)} \iint M_y(z) \tag{14}$$

$$\Delta_{x2} = \iint \frac{d^2x}{dz^2} = \frac{I_{xx}}{E(I_{xx}I_{yy} - I_{xy}^2)} \iint M_y(z) \tag{15}$$

Re-arranging Equations 14 and 15 gives

$$\iint M_x(z) = \frac{E\left(I_{xx}I_{yy} - I_{xy}^2\right)}{-I_{xy}}\Delta_{y2} \tag{16}$$

$$\iint M_x(z) = \frac{E\left(I_{xx}I_{yy} - I_{xy}^2\right)}{I_{xx}}\Delta_{x2} \tag{17}$$

Setting Equation 16 equal to Equation 17 and simplifying gives

$$\Delta_{y2} = -\frac{I_{xy}}{I_{xx}}\Delta_{x2} \tag{18}$$

Equation 18 gives the vertical deflection of an asymmetric section as a function of lateral deflection.

Superimposing the deflections from Case 1 and Case 2 gives the total deflection

$$\Delta_x = \Delta_{x1} + \Delta_{x2} \tag{19}$$

$$\Delta_y = \Delta_{y1} + \Delta_{y2} \tag{20}$$

Lateral Deflections - Evaluation

Lateral deflections will be evaluated for a long span deck girder bridge with narrow top flanges. The narrow top flanges result in a narrow girder spacing as is typical for long span bridges. The narrow top flanges also represent deck girders with smaller lateral stiffness, and hence larger lateral deflections, then deck girders with wider top flanges. The bridge to be evaluated is taken from the WSDOT span capability analysis for the WF69DG girder. The bridge properties are as follows

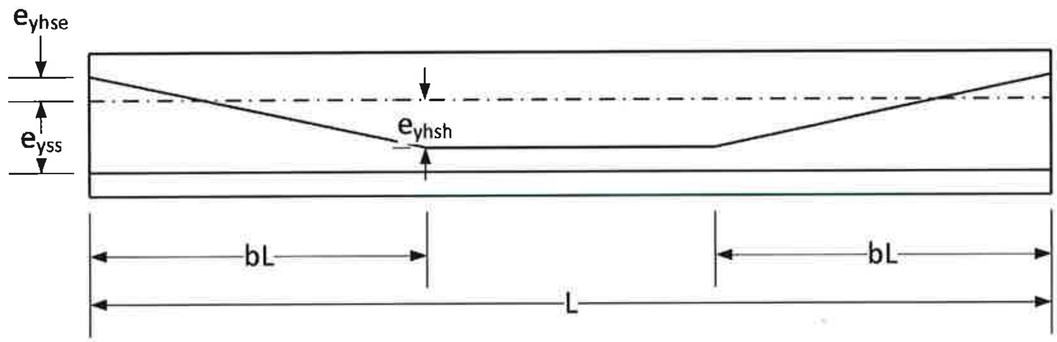
Girders	6 – WF69DG, 5 ft top flange
Longitudinal Joint	9" UHPC
Bridge Length	200 ft
CL Brg – CL Brg span length	197 ft

Bridge sections for 2% normal crown and 4% and 6% full superelevation will be evaluated for the three typical construction methods.

Concrete Properties

$$f'_{ci} = 7.5\ ksi$$
$$E_{ci} = 5605.5\ ksi$$
$$\gamma_c = 0.165\ kft$$

Prestressing



Straight Strands	Harped Strands
$P_{ss} = 1832.61\ kip$ $e_{y_{ss}} = 37.462\ in$	$P_{hs} = 956.14\ kip$ $e_{y_{hse}} = -16.051 in$ $e_{y_{hsh}} = 36.049\ in$

The lateral deflections for Girder C will be computed.

Permissible Lateral Fabrication Sweep

Lateral sweep due to inaccuracies in fabrication are generally small. However, WSDOT permits lateral sweep up to 1/8" per 10 ft of girder length, not to exceed ½". For the subject bridge, the maximum permissible lateral sweep is

$$\Delta_{sweep} = \overset{Min}{\cancel{Max}}\left[\left(\frac{1}{8}in\right)\left(\frac{197ft}{10ft}\right), 0.5in\right] = \overset{Min}{\cancel{Max}}(2.46\ in, 0.5\ \text{in}) = 0.5\ in$$

Configuration 1

Girders have equal length top flange overhangs. Girder installed with webs normal to roadway surface.

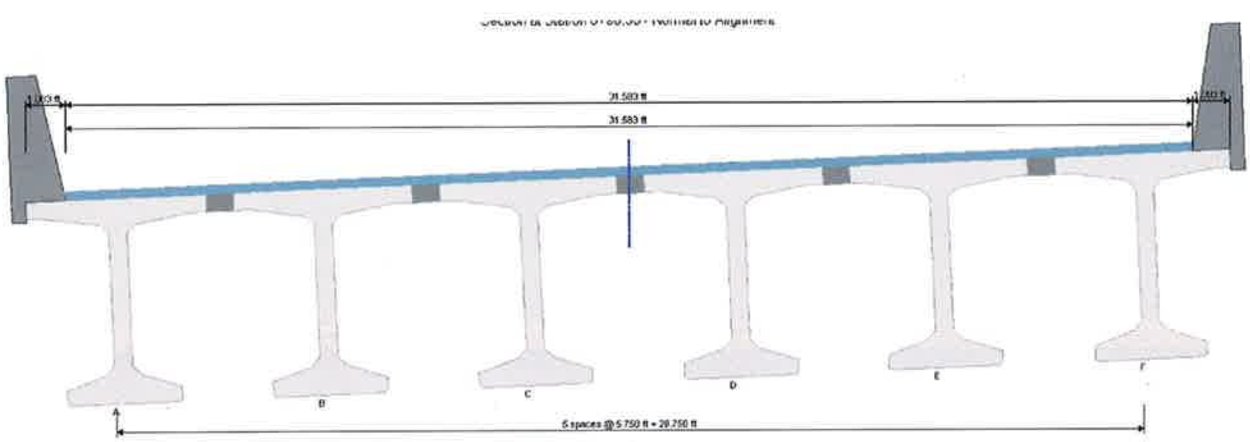


Figure 5 - Configuration 1 bridge section

Girder Properties

$$A = 1087.5in^2$$
$$I_{xx} = 769358.2in^4$$
$$I_{yy} = 150453.0in^4$$
$$I_{xy} = 0in^4$$
$$\overline{y_t} = 30.296in$$
$$\overline{y_b} = 38.704in$$
$$w = 1.246\frac{k}{ft}$$

We are concerned with the fit up of the girder top flanges after erection. We are only concerned about deflections that alter the straightness and width of the joint, therefore the only the weak axis deflections are of interest for this bridge configuration.

For small angles $\sin(\theta) = \theta, \cos(\theta) = 1$

$$w_x = w\sin(\theta) = w\theta$$

$$\Delta_x = -\frac{5w_xL^4}{384E_{ci}I_{yy}} = -\frac{5w\theta L^4}{384E_{ci}I_{yy}}$$

Should be using E_c

	2%	4%	6%
$\Delta_x\ (in)$	−1.00	−2.00	−3.00

At the full superelevation, all girders will deflection the same amount and in the same direction. The joint will not be straight; however, it will maintain a constant width.

At the 2% normal crown slope, shown in Figure 6, the three girders on the left will deflect toward the left and the three girders on the right will deflect toward the right. This will result in a joint between Girders C and D that is 2" wider at mid-span than at the ends of the bridge. If the maximum permissible lateral sweep due to fabrication inaccuracies occurs in both girders the joint width could be as much as 3" wider than designed. Loss of non-contact splice overlap is of concern.

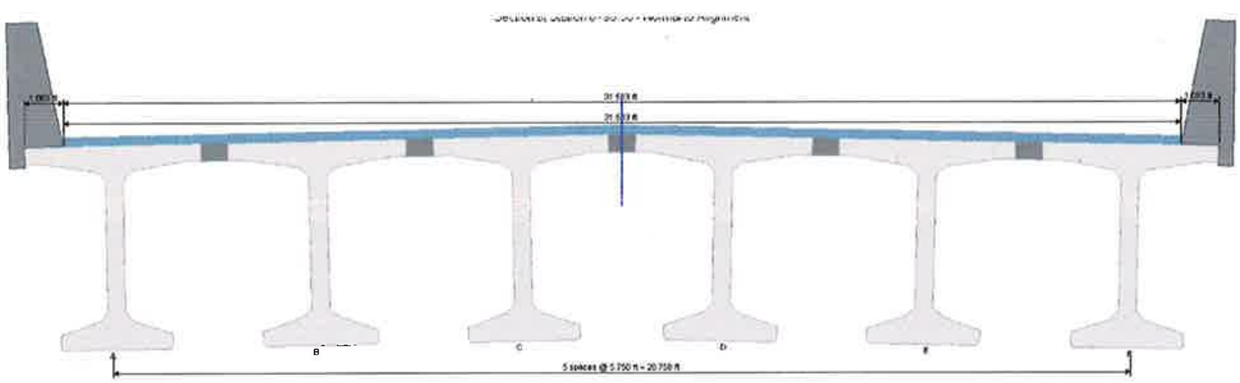


Figure 6 - Configuration 1 at normal crown section

Configuration 2

Girders have equal length top flange overhangs. Girders installed with webs plumb. The top flange is thickened transversely to accommodate the roadway cross slope and superelevation.

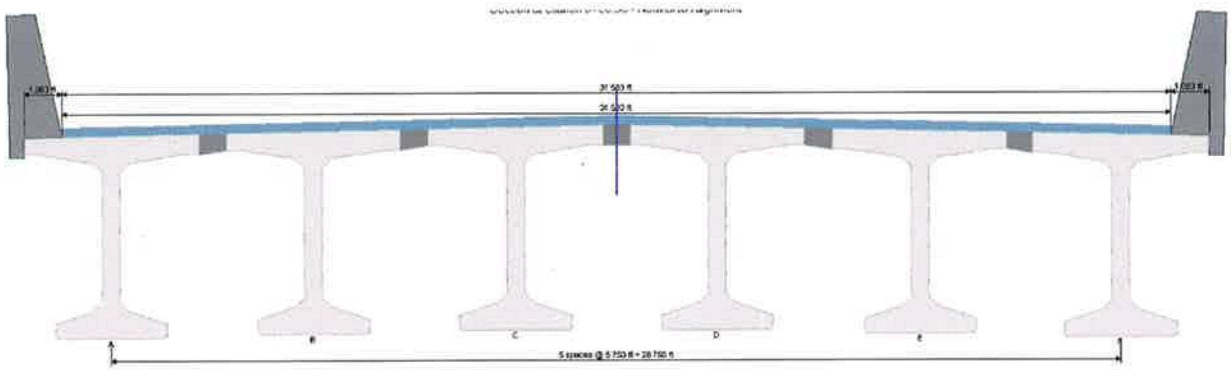


Figure 7 - Configuration 2

Girder Properties

	2%	4%	6%
$A(in^2)$	1123.531	1159.531	1195.531
$I_{xx}(in^4)$	802195.3	834679.9	866894.2
$I_{yy}(in^4)$	161137.7	171605.9	181877.4
$I_{xy}(in^4)$	10768.5	21287.0	31590.9
$\overline{x_l}(in)$	30.32	30.621	30.903
$\overline{x_r}(in)$	29.68	29.379	29.097
$\overline{y_t}(in)$	30.513	30.765	31.051
$\overline{y_b}(in)$	39.687	40.635	41.549
$w\left(\frac{k}{ft}\right)$	1.287	1.329	1.370
$\frac{V}{S}(in)$	3.307	3.805	3.907

Detailed computations for 6% superelevation

Girder self-weight deflections

$$\Delta_{y1} = -\frac{5wL^4}{384E_{ci}}\frac{I_{yy}}{(I_{xx}I_{yy} - I_{xy}^2)}$$

$$\Delta_{x1} = -\frac{I_{xy}}{I_{yy}}\Delta_{y1}$$

$$\Delta_{y1} = -\frac{5\left(1.370\frac{k}{ft}\right)(197ft)^4}{384(5605.5ksi)}\left(\frac{1728in^3}{1\,ft^3}\right)\left(\frac{181877.4in^4}{(866894.2in^4)(181877.4in^4) - (31590.9in^4)^2}\right)$$
$$= -9.615in$$

$$\Delta_{x1} = -\left(\frac{31590.9in^4}{181877.4in^4}\right)(-9.615in) = 1.670\ in$$

$$\Delta_{y2} = 0in$$

$$\Delta_{x2} = 0\ in$$

$$\Delta_{y\_gdr} = -9.615 + 0 = -9.615in$$

$$\Delta_{x\_gdr} = 1.670 + 0 = 1.670in$$

Deflections due to prestressing

The prestressing is installed symmetrically with respect to the CL Web. However, the web is offset from the CG of the section. This will result in lateral deflection due to prestressing.

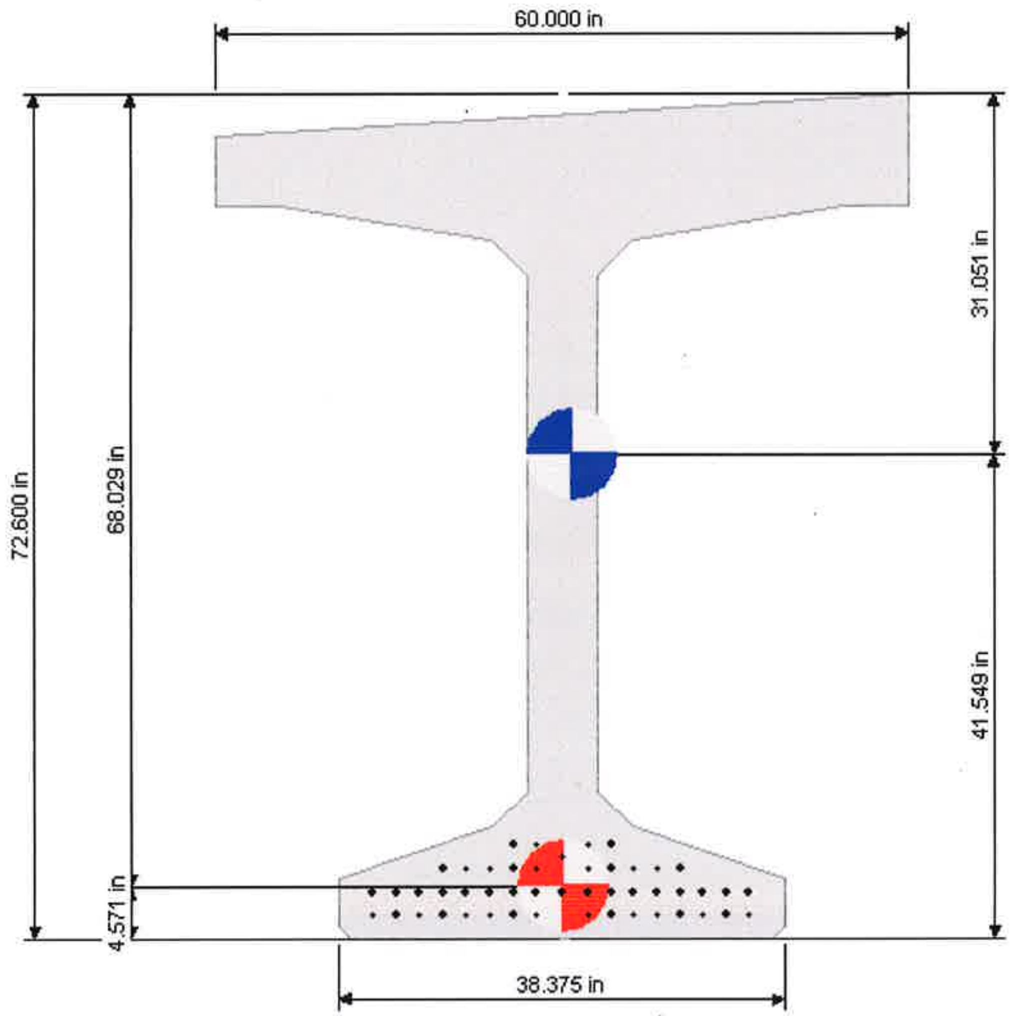


Figure 8 - Girder section showing CG prestressing (red circle) laterally offset from the CG of the girder (blue circle)

Lateral strand eccentricity for straight and harped strands

$$e_x = 30.903in - 30.000in = 0.903in$$

Equivalent prestress forces

Straight Strands

$$M_{x_{ss}} = P_{ss}e_{y_{ss}} = (1832.61kip)(37.462in) = 5721.09\ k-ft$$

$$M_{y_{ss}} = P_{ss}e_x = (1832.61kip)(0.903in) = 137.90\ k-ft$$

Harped Strands

$$M_{x_{hs}} = P_{hs}e_{y_{hse}} = (956.14kip)(-16.051in) = -1278.92\ k-ft$$

$$M_{y_{hs}} = P_{hs}e_x = (956.14kip)(0.903in) = 71.95\ k-ft$$

$$e' = e_{y_{hsh}} - e_{y_{hse}} = 36.049 + 16.051 = 52.1\ in$$

$$N = \frac{P_{hs}e'}{bL} = \frac{(956.14kip)(52.1in)}{0.4(197ft)}\left(\frac{1ft}{12in}\right) = 52.68\ kip$$

Deflections

$$\Delta_{y1} = \frac{(M_{x_{ss}} + M_{x_{hs}})L^2}{8E_{ci}}\frac{I_{yy}}{I_{xx}I_{yy} - I_{xy}^2} + \frac{b(3-4b^2)NL^3}{24E_{ci}}\frac{I_{yy}}{I_{xx}I_{yy} - I_{xy}^2}$$
$$= \frac{(5721.09 - 1278.92\ k-ft)(197ft)^2}{8(5605.5ksi)}\left(\frac{1728in^3}{1ft^3}\right)\left(\frac{181877.4in^4}{(866894.2in^4)(181877.4in^4) - (31590.9in^4)^2}\right)$$
$$+ \frac{0.4(3-(4)(0.4)^2)(52.68kip)(197ft)^3}{24(5605.5ksi)}\left(\frac{1728in^3}{1ft^3}\right)\left(\frac{181877.4in^4}{(866894.2in^4)(181877.4in^4) - (31590.9in^4)^2}\right)$$
$$= 7.712in + 5.669in = 13.381in$$

$$\Delta_{x1} = \left(-\frac{I_{xy}}{I_{yy}}\right)\Delta_{y1} = \left(-\frac{31590.9in^4}{181877.4in^4}\right)(13.381in) = -2.324in$$

$$\Delta_{x2} = \frac{(M_{y_{ss}} + M_{y_{hs}})L^2}{8E_{ci}}\frac{I_{xx}}{I_{xx}I_{yy} - I_{xy}^2}$$
$$= \frac{(137.9 + 71.95k-ft)(197ft)^2}{8(5605.5ksi)}\left(\frac{1728in^3}{1ft^3}\right)\left(\frac{866894.2in^4}{(866894.2in^4)(181877.4in^4) - (31590.9in^4)^2}\right)$$
$$= 1.736in$$

$$\Delta_{y2} = \left(-\frac{I_{xy}}{I_{xx}}\right)\Delta_{x2} = \left(-\frac{31590.9in^4}{866894.2in^4}\right)(1.736in) = -0.063in$$

Total prestress deflection

$$\Delta_{y\_ps} = 13.381in - 0.063in = 13.318in$$

$$\Delta_{x\_ps} = -2.324in + 1.736in = -0.588in$$

Initial deflection

$$\Delta_y = \Delta_{y\_gdr} + \Delta_{y\_ps} = -9.615in + 13.318in = 3.703in$$

$$\Delta_x = \Delta_{x\_gdr} + \Delta_{x\_ps} = 1.670in - 0.588in = 1.082in$$

Creep coefficient

$$\psi(t, t_i) = 1.9 k_s k_{hc} k_f k_{td} t_i^{-0.118}$$

$$k_s = 1.45 - 0.13\left(\frac{V}{S}\right) \geq 1.0 = 1.45 - 0.13(3.907) = 0.942 \rightarrow 1.00$$

$$k_{hc} = 1.56 - 0.008H = 1.56 - 0.008(75) = 0.96$$

$$k_f = \frac{5}{1 + f'_{ci}} = \frac{5}{1 + 7.5} = 0.588$$

$$k_{td} = \frac{t}{12\left(\frac{100 - 4f'_{ci}}{f'_{ci} + 20}\right) + t} = \frac{119}{12\left(\frac{100 - 4(7.5)}{7.5 + 20}\right) + 119} = 0.796$$

$$\psi(119,1) = 1.9(1.00)(0.96)(0.588)(0.796)(1)^{-0.118} = 0.854$$

Deflections at the time of erection (120 days)

$$\Delta_y = (1 + 0.854)(3.703in) = 6.865in$$

$$\Delta_x = (1 + 0.854)(1.082in) = 2.006in$$

Deflection summary

	2%	4%	6%
	GIRDER DEFLECTION (IN)		
Δ_{y1}	-9.710	-9.654	-9.614
Δ_{y2}	0	0	0
Δ_{x1}	0	0	0
Δ_{x2}	0.649	1.197	1.670
Δ_y	-9.711	-9.654	-9.614
Δ_x	0.649	1.197	1.670
	PRESTRESS DEFLECTION (IN)		
Δ_{y1}	13.600	13.483	13.381
Δ_{y2}	-0.009	-0.032	-0.063
Δ_{x1}	-0.909	-1.673	-2.324
Δ_{x2}	0.689	1.259	1.737
Δ_y	13.591	13.451	13.318
Δ_x	-0.220	-0.413	-0.587
	INITIAL DEFLECTION (IN)		
Δ_y	3.381	3.797	3.704

Δ_x	0.429	0.784	1.083
	DEFLECTION AT ERECTION (IN)		
Δ_y	7.194	7.039	6.866
Δ_x	0.796	1.454	2.007

The lateral deflections for this configuration are improved over Configuration 1. The deflections now toward the bridge centerline instead of away from it. For the 2% normal crown slope case, the joint will be 1.6" narrower, or approximately 2.6" narrower, if maximum permissible fabrication sweep occurs, at mid-span then at the ends. Loss of development length is of concern.

Configuration 3

Girders have unequal length top flange overhangs such that the CG of the girder section aligns with the CL Web. Girders installed with webs plumb. The top flange is thickened transversely to accommodate the roadway cross slope and superelevation. Figure 9 shows a WF69DG girder with the top flange overhang lengths as a function of the parameter k.

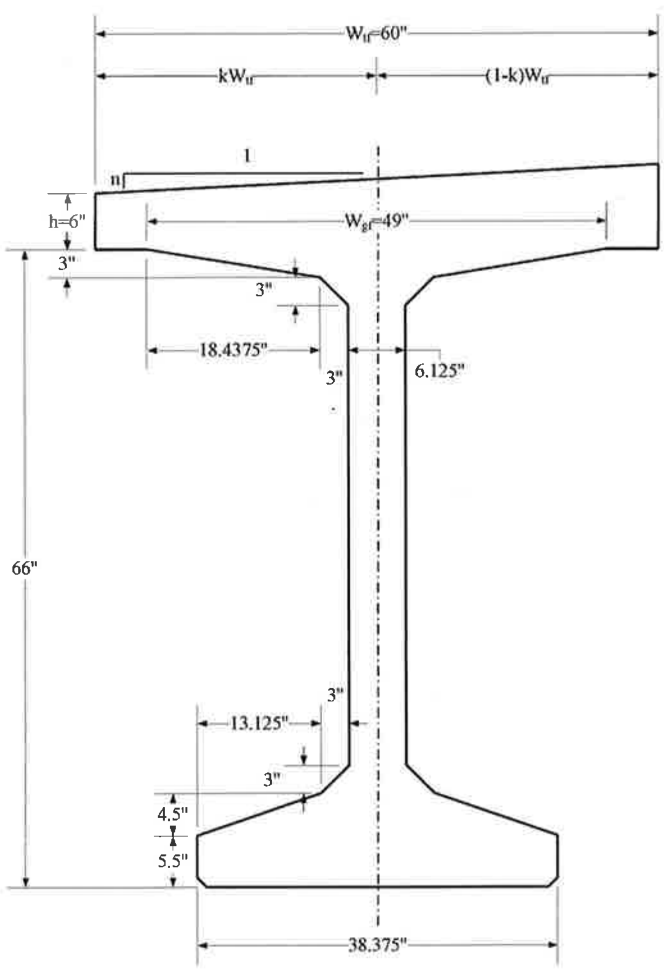


Figure 9 - WF69DG girder dimensions

The center of gravity of the section is on the centerline of the web when $k = \frac{1}{3}\frac{3h+2nW}{2h+nW}$.

For a 6% cross slope,

$$k = \left(\frac{1}{3}\right)\frac{3(6in) + 2(0.06)(60in)}{2(6in) + (0.06)(60in)} = 0.53846$$

The length of the left overhang is $kW = (0.53846)(60in) = 32.308in$ and the length of the right overhang is $(1-k)W = (1-0.53846)(60in) = 27.692in$

Girder Properties

	2%	4%	6%
$A(in^2)$	1123.531	1159.531	1195.531
$I_{xx}(in^4)$	802195.3	834679.9	866894.2
$I_{yy}(in^4)$	160925.7	170853.0	180360.7
$I_{xy}(in^4)$	1184.7	2568.0	4137.2
$\overline{x_l}(in)$	30.909	31.667	32.308
$\overline{x_r}(in)$	29.091	28.333	27.692
$\overline{y_t}(in)$	30.513	30.765	31.051
$\overline{y_b}(in)$	39.687	40.635	41.549
$w\left(\frac{k}{ft}\right)$	1.287	1.329	1.370
$\frac{V}{S}(in)$	3.307	3.805	3.907

Detailed computations for 6% superelevation

Girder self-weight deflections

$$\Delta_{y1} = -\frac{5wL^4}{384E_{ci}}\frac{I_{yy}}{I_{xx}I_{yy} - I_{xy}^2}$$

$$\Delta_{x1} = -\frac{I_{xy}}{I_{yy}}\Delta_{y1}$$

$$\Delta_{y1} = -\frac{5\left(1.370\frac{k}{ft}\right)(197ft)^4}{384(5605.5ksi)}\left(\frac{1728in^3}{1\,ft^3}\right)\left(\frac{180360.7in^4}{(866894.2in^4)(180360.7in^4) - (4137.2in^4)^2}\right)$$
$$= -9.554in$$

$$\Delta_{x1} = -\left(\frac{4137.2in^4}{180360.7in^4}\right)(-9.554in) = 0.219\ in$$

$$\Delta_{y2} = 0in$$

$$\Delta_{x2} = 0\ in$$

$$\Delta_{y\_gdr} = -9.554 + 0 = -9.554in$$

$$\Delta_{x\_gdr} = 0.219 + 0 = 0.219in$$

Deflections due to prestressing

The prestressing is installed symmetrical with respect to the CL Web. The top flange overhang dimensions have been computed so that the CG of the section aligned with the CL Web. The prestressing will not cause lateral deflection.

Lateral strand eccentricity for straight and harped strands

$$e_x = 30.903 - 30.903 = 0.0in$$

Equivalent prestress forces

Straight Strands

$$M_{x_{ss}} = P_{ss}e_{y_{ss}} = (1832.61kip)(37.462in) = 5721.09\ k - ft$$

$$M_{y_{ss}} = P_{ss}e_x = (1832.61kip)(0.0in) = 0\ k - ft$$

Harped Strands

$$M_{x_{hs}} = P_{hs}e_{y_{hse}} = (956.14kip)(-16.051in) = -1278.92\ k - ft$$

$$M_{y_{hs}} = P_{hs}e_x = (956.14kip)(0.0in) = 0\ k - ft$$

$$e' = e_{y_{hsh}} - e_{y_{hse}} = 36.049 + 16.051 = 52.1\ in$$

$$N = \frac{P_{hs}e'}{bL} = \frac{(956.14kip)(52.1in)}{0.4(197ft)}\left(\frac{1ft}{12in}\right) = 52.68\ kip$$

Prestress deflections

$$\Delta_{y1} = \frac{(M_{x_{ss}} + M_{x_{hs}})L^2}{8E_{ci}}\frac{I_{yy}}{I_{xx}I_{yy} - I_{xy}^2} + \frac{b(3-4b^2)NL^3}{24E_{ci}}\frac{I_{yy}}{I_{xx}I_{yy} - I_{xy}^2}$$
$$= \frac{(5721.09 - 1278.92k - ft)(197ft)^2}{8(5605.5ksi)}\left(\frac{1728in^3}{1ft^3}\right)\left(\frac{180360.7in^4}{(866894.2in^4)(180360.7in^4) - (4137.2in^4)^2}\right)$$
$$+ \frac{0.4(3-(4)(0.4)^2)(52.68kip)(197ft)^3}{24(5605.5ksi)}\left(\frac{1728in^3}{1ft^3}\right)\left(\frac{180360.7in^4}{(866894.2in^4)(180360.7in^4) - (4137.2in^4)^2}\right)$$
$$= 7.664in + 5.634in = 13.298in$$

$$\Delta_{x1} = \left(-\frac{I_{xy}}{I_{yy}}\right)\Delta_{y1} = \left(-\frac{4137.2in^4}{180360.7in^4}\right)(13.298in) = -0.305in$$

$$\Delta_{x2}=\frac{(M_{y_{ss}}+M_{y_{hs}})L^2}{8E_{ci}}\frac{I_{xx}}{I_{xx}I_{yy}-I_{xy}^2}$$
$$=\frac{(0k-ft)(197ft)^2}{8(5605.5ksi)}\left(\frac{1728in^3}{1ft^3}\right)\left(\frac{866894.2in^4}{(866894.2in^4)(180360.7in^4)-(4137.2in^4)^2}\right)$$
$$=0in$$

$$\Delta_{y2}=\left(-\frac{I_{xy}}{I_{xx}}\right)\Delta_{x2}=\left(-\frac{4137.2in^4}{866894.2in^4}\right)(0in)=0in$$

Total prestress deflection

$$\Delta_{y\_ps}=13.298in-0.0in=13.298in$$

$$\Delta_{x\_ps}=-0.305in+0in=-0.305in$$

Initial deflection

$$\Delta_y=\Delta_{y\_gdr}+\Delta_{y\_ps}=-9.554in+13.298in=3.744in$$

$$\Delta_x=\Delta_{x\_gdr}+\Delta_{x\_ps}=0.219in-0.305in=-0.086in$$

Creep

$$\psi(t,t_i)=1.9k_sk_{hc}k_fk_{td}t_i^{-0.118}$$

$$k_s=1.45-0.13\left(\frac{V}{S}\right)\geq 1.0=1.45-0.13(3.907)=0.942\rightarrow 1.00$$

$$k_{hc}=1.56-0.008H=1.56-0.008(75)=0.96$$

$$k_f=\frac{5}{1+f'_{ci}}=\frac{5}{1+7.5}=0.588$$

$$k_{td}=\frac{t}{12\left(\frac{100-4f'_{ci}}{f'_{ci}+20}\right)+t}=\frac{119}{12\left(\frac{100-4(7.5)}{7.5+20}\right)+119}=0.796$$

$$\psi(119,1)=1.9(1.00)(0.96)(0.588)(0.796)(1)^{-0.118}=0.854$$

Deflections at the time of erection (120 days)

$$\Delta_y=(1+0.854)(3.744in)=6.940in$$

$$\Delta_x=(1+0.854)(-0.086in)=-0.159in$$

Deflection summary

	2%	4%	6%
		GIRDER DEFLECTION (IN)	
Δ_{y1}	1.287	1.329	1.370
Δ_{y2}	-9.702	-9.624	-9.554

Δ_{x1}	0.000	0.000	0.000
Δ_{x2}	0.000	0.000	0.000
Δ_y	0.071	0.145	0.219
Δ_x	-9.702	-9.624	-9.554
	PRESTRESS DEFLECTION (IN)		
Δ_{y1}	13.588	13.441	13.298
Δ_{y2}	0.000	0.000	0.000
Δ_{x1}	-0.100	-0.202	-0.305
Δ_{x2}	0.000	0.000	0.000
Δ_y	13.588	13.441	13.298
Δ_x	-0.100	-0.202	-0.305
	INITIAL DEFLECTION (IN)		
Δ_y	3.886	3.818	3.744
Δ_x	-0.029	-0.057	-0.086
	DEFLECTION AT ERECTION (IN)		
Δ_y	7.204	7.077	6.940
Δ_x	-0.053	-0.106	-0.159

The lateral deflections are significantly smaller then for Configurations 1 and 2. The deflections are away from the bridge centerline. The magnitudes of the deflections are small. The joints will remain essentially straight, though maximum permissible fabrication sweep may alter the joint width by approximately 1". Development and non-contact splice overlap are not of concern.

Conclusions

UHPC joints rely on proper top flange alignment and joint fit up. This study shows that attention to lateral deflections of precast deck girders is important to ensuring good joint construction. All three construction methods are viable when lateral deflections are within manageable limits. The engineer of record should evaluate lateral deflections and assess its impact on longitudinal joint fit up and performance.